A c_0-detector criterion for the Banakh–Cauty linear metric AR problem

Candidate partial-result packet for arXiv:1006.3092

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Status. Candidate partial result; likely valid, subject to expert review. This packet does not claim to settle either source problem in full.

1 Source question and result

Banakh and Cauty ask in Section 4, page 8 of [1]:

> *Is each complete linear metric AR-space homeomorphic to a Hilbert space?*

They also ask the stronger convex-subset version displayed in the source crop.

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4. OPEN PROBLEMS

The proof of Theorem 5 heavily exploits the machinery of Banach space theory and does not work in the non-locally convex case. This leaves the following problem open:

Problem 1. *Is each non-separable completely metrizable convex AR-subset of a linear metric space homeomorphic to a Hilbert space?*

Even a weaker problem seems to be open:

Problem 2. *Is each complete linear metric AR-space homeomorphic to a Hilbert space?*

This is true in the separable case, see [4], [5].

Here a continuous linear map $T : X \to c_0$ is called *compact* if $T(V)$ is totally bounded for some neighbourhood V of zero. This is the definition used by Banakh–Cauty. The main result of this packet is the following sufficient condition.

Theorem 1 (noncompact c_0 detector)**.** *Let X be a complete linear metric absolute retract. If there is a continuous noncompact linear operator*

$$T : X \longrightarrow c_0,$$

then X is homeomorphic to the Hilbert space $\ell_2(\operatorname{dens} X)$.

This removes local convexity from the source theorem whenever such a detector exists. For quasi-Banach spaces it gives a particularly transparent partial answer.

Corollary 2 (Banach-envelope subcase)**.** *Let X be a quasi-Banach space which, as a topological space, is an absolute retract. If its Banach envelope $\widehat{X}$ is infinite-dimensional, then $X \cong \ell_2(\operatorname{dens} X)$.*

If instead $\widehat{X}$ is finite-dimensional and X is infinite-dimensional, then

$$X \cong N \times \widehat{X}$$

linearly and topologically, where N is a complete linear metric AR-space with trivial continuous dual and trivial Banach envelope. Thus, within the quasi-Banach class, the part not reached by the theorem reduces to the trivial-envelope case.

2 Proof intuition

The source proof uses local convexity twice. The first use is to replace the identity by a map whose image is locally contained in a finite convex hull. For an AR this can instead be done topologically: embed X as a closed subset of a Banach space, take a cover-controlled barycentric approximation there, and retract it to X. The resulting map is joined to the identity by a small homotopy and has locally relatively compact image.

The second part of the source argument is linear rather than locally convex. Noncompactness of T supplies small markers whose c_0 images have a large new coordinate and tiny later-coordinate interactions with all earlier markers. A two-marker partition of unity inserts these markers into the locally precompact approximation. For two distinct inserted codes, one of five exhaustive coefficient/order cases exposes a coordinate separated from zero. This gives the separated approximation property (SAP), hence LFAP by Banakh–Cauty, and their Hilbert-space recognition theorem finishes the proof.

The original problem remains open for spaces with no such linear detector. For a quasi-Banach space, every map to a Banach space factors through its Banach envelope; consequently the trivial-envelope class is a genuine hard boundary of this method, not merely a missing estimate.

3 Proof

We use two lemmas. The first is the point at which the AR hypothesis replaces local convexity.

Lemma 3 (controlled locally precompact approximation)**.** *Let (X, d) be a completely metrizable AR, where d is any compatible metric. For every $\varepsilon > 0$ there are a map $f : X \to X$ and a homotopy H from id_X to f such that every track $H(\{x\} \times [0, 1])$ has d-diameter less than ε, and every $x \in X$ has a neighbourhood O_x for which $f(O_x)$ is relatively compact.*

Proof. Choose a bounded complete compatible metric on X and use the Kuratowski embedding to identify X with a closed subset of a Banach space E. Since X is an AR, there is a retraction $r : E \to X$.

For each $x \in X$, continuity of r gives a convex norm-ball W_x about x such that $r(W_x)$ has d-diameter less than ε. Take a locally finite star-refinement $\mathcal{V}$ of the cover $\{W_x \cap X : x \in X\}$, choose $x_V \in V$ for every $V \in \mathcal{V}$, and let $(\lambda_V)_{V\in\mathcal{V}}$ be a subordinate partition of unity. Put

$$g(y) = \sum_{V\in\mathcal{V}} \lambda_V(y)x_V \quad \text{in } E, \qquad f = r \circ g,$$

and

$$H(y, t) = r\big((1 - t)y + tg(y)\big).$$

If $\lambda_V(y) > 0$, all the participating V's meet at y; the star-refinement therefore puts y, all corresponding x_V, their convex hull, and the joining segment inside a single W_x. Hence the homotopy has the required control. Local finiteness gives a neighbourhood O_y on which only finitely many λ_V occur. Thus $g(O_y)$ lies in the convex hull of finitely many points, whose closure is compact, and $f(O_y)$ is relatively compact. $\square$

The next lemma isolates and audits the marker part of the argument in [1, proof of Lemma 2]. We include the details because the published formulas contain harmless index slips and because no local convexity is used here.

Lemma 4 (c_0 marker lemma)**.** *Let (X, d) be a linear metric space. Suppose that*

1. *for every $\varepsilon > 0$, the identity has an ε-small homotopic approximation $f : X \to X$ which is locally relatively compact; and*
2. *there is a continuous noncompact linear map $T : X \to c_0$.*

Then (X, d) has SAP.

Proof. Fix $\varepsilon > 0$. Choose a balanced zero-neighbourhood B so small that $B + B$ has d-diameter less than $\varepsilon/3$, and, shrinking B if necessary, assume $T(B)$ is contained in the unit ball of c_0. Since T is noncompact, $T(B)$ is not totally bounded. The usual compactness criterion in c_0 therefore supplies $0 < \delta \le 1$ such that

$$\text{for every } N \text{ there are } u \in B, q \ge N \text{ with } |(Tu)_q| \ge \delta. \tag{1}$$

Put $\alpha = \delta^3/100$. Inductively choose $u_r \in B$ and strictly increasing coordinates q_r such that

$$|(Tu_r)_{q_r}| \ge \delta, qquad |(Tu_s)_{q_r}| < \alpha \quad (s < r). \tag{2}$$

Indeed, after $u_0, \dots, u_{r-1}$ have been chosen, their finitely many c_0 tails are uniformly below α, and (1) chooses the next coordinate and marker.

Let $\xi : \mathbb{N}^2 \to \mathbb{N}$ be injective, increasing in its first argument, and satisfying $\xi(i, k) > i$. This is obtained by dividing $\mathbb{N}$ into countably many infinite sets. Write

$$u_{i,k} = u_{\xi(i,k)}, \qquad \phi_{i,k} = e^*_{q_{\xi(i,k)}} T.$$

Thus $|\phi_{i,k}(u_{i,k})| \ge \delta$; a functional belonging to a later ξ-rank is smaller than α on every earlier marker; and all marker evaluations have modulus at most one.

Apply Lemma 3, with control $\varepsilon/3$, to obtain f. For $s \in \mathbb{N}$, let C_s be the open set of points having a neighbourhood O on which

$$|e^*_{q_r} T f(y)| < \alpha \quad (y \in O,\ r \ge s).$$

Local relative compactness of f and uniform tail decay on compact subsets of c_0 show that the increasing open sets C_s cover X.

A standard shrinking of this increasing cover gives increasing open sets $U_0 = \varnothing, U_1, \dots$ covering X with $\overline{U}_{i+1} \subset C_i \cap U_{i+2}$. Take a partition of unity (λ_i) subordinate to

$$U_{i+1} \setminus \overline{U}_{i-1} \quad (U_{-1} = U_0 = \varnothing).$$

If $x \in U_{i+1} \setminus U_i$, exactly the two terms $\lambda_i(x)$ and $\lambda_{i+1}(x)$ can be nonzero, and their sum is one. Define

$$f_k(x) = f(x) + \sum_i \lambda_i(x) u_{i,k}. \tag{3}$$

The sum is locally finite and in fact has at most two terms. The straight homotopy from f to f_k, concatenated with the homotopy from id_X to f, is an ε-homotopy: every difference along its second part is in $B + B$.

It remains to prove uniform separation. Fix $k \neq n$ and points $x \in U_{i+1} \setminus U_i$, $y \in U_{j+1} \setminus U_j$. Put

$$a_0 = \xi(i,k), \quad a_1 = \xi(i+1,k), \quad b_0 = \xi(j,n), \quad b_1 = \xi(j+1,n),$$

and write $s = \lambda_{i+1}(x)$, $t = \lambda_{j+1}(y)$. The four ranks are distinct, $a_0 < a_1$, $b_0 < b_1$, and we may interchange the two points so that $a_1 < b_1$. At any coordinate selected below, its rank is at least $\max\{i,j\}$; hence the two base terms $Tf(x), Tf(y)$ each contribute less than α.

The following five cases are exhaustive. In the indicated coordinate, the displayed lower bound follows from (2); an "early-marker" error costs α, while a later marker costs at most its coefficient.

1. If $t > \delta^2/10$, use rank b_1 and obtain $t\delta - 5\alpha > \delta^3/20$.
2. If $t \leq \delta^2/10$ and $b_0 > a_1$, use b_0 and obtain $(1-t)\delta - t - 4\alpha > \delta^3/20$.
3. If $t \leq \delta^2/10$, $b_0 < a_1$, and $s > \delta/4$, use a_1 and obtain $s\delta - t - 4\alpha > \delta^3/20$.
4. If $t \leq \delta^2/10$, $b_0 < a_1$, $s \leq \delta/4$, and $a_0 < b_0$, use b_0 and obtain $(1-t)\delta - s - t - 3\alpha > \delta^3/20$.
5. In the remaining case $a_0 > b_0$, use a_0 and obtain $(1-s)\delta - s - t - 3\alpha > \delta^3/20$.

For completeness, the inequalities use only $0 < \delta \leq 1$, $\alpha = \delta^3/100$, $t \leq \delta^2/10$, and $s \leq \delta/4$ where those hypotheses occur. Thus

$$\|T(f_k(x) - f_n(y))\|_\infty > \delta^3/20$$

in every case. By continuity of T, some $\eta > 0$, independent of x, y, k, n, satisfies

$$d(f_k(x), f_n(y)) \geq \eta \qquad (k \neq n).$$

The images $f_k(X)$ are separated, proving SAP. □

Proof of Theorem 1. Lemma 3 and Lemma 4 give SAP. Banakh and Cauty prove that every metric space with SAP has LFAP [1, Lemma 1]. Their Banakh–Zarichnyy criterion states that a convex subset of a linear metric space is homeomorphic to an infinite-dimensional Hilbert space exactly when it is a completely metrizable AR with LFAP [1, Theorem 4]. Apply it to the linear space X, viewed as a convex subset of itself. A noncompact operator cannot have finite-dimensional domain, and density is a topological invariant, so the Hilbert space is $\ell_2(\operatorname{dens} X)$. □

Proof of Corollary 2. Let $J : X \to \widehat{X}$ be the canonical envelope map. If $\widehat{X}$ is infinite-dimensional, the Josefson–Nissenzweig theorem [2] gives norm-one functionals $\psi_m \in \widehat{X}^*$ converging weak-star to zero. Hence

$$T(x) = (\psi_m(Jx))_{m \in \mathbb{N}}$$

defines a continuous linear map $X \to c_0$. It is noncompact. Indeed, the envelope unit ball is the closed convex hull of $J(B_X)$, so $\|\psi_m\| = 1$ implies $\sup_{x \in B_X} |\psi_m(Jx)| = 1$. Therefore $T(B_X)$ has no uniformly vanishing tails and is not totally bounded. If the image of any zero-neighbourhood were totally bounded, scaling a quasi-norm ball into that neighbourhood would make $T(B_X)$ totally bounded as well. Thus T is noncompact, and Theorem 1 applies.

Now suppose $\widehat{X}$ is finite-dimensional. The dense linear subspace $J(X)$ equals $\widehat{X}$. Choose preimages of a basis to obtain a continuous linear section $S : \widehat{X} \to X$. Thus $X = N \oplus S(\widehat{X})$, where $N = \ker J$ is closed and complemented. It is complete and, as a retract of the AR X, is an AR. Every continuous functional on N would extend to X through the continuous projection onto N, but all functionals on X vanish on $\ker J$; hence $N^* = \{0\}$ and its Banach envelope is trivial. □

4 Upgrade attempts and obstruction audit

Five materially different upgrade steps were pursued.

1. The source's locally convex barycentric approximation was replaced by the AR retraction construction of Lemma 3. This succeeded.

2. The source marker proof was re-derived with only a noncompact $T : X \to c_0$ and audited by the five exhaustive cases above. This succeeded.

3. Josefson–Nissenzweig was applied to the Banach envelope. This upgrades the theorem to every quasi-Banach AR with infinite-dimensional envelope.

4. A finite-dimensional envelope was split off. This reduces the remaining quasi-Banach case to a complemented AR with trivial dual and trivial envelope.

5. Nonlinear c_0 detectors and purely metric separated markers were tested as routes through the last case. The separation calculation needs additivity to isolate a marker from the uncontrolled base term $f(x) - f(y)$. Moreover every continuous linear map from a trivial-envelope quasi-Banach space to c_0 is zero by the envelope universal property. Thus the strongest remaining route would require a genuinely nonlinear replacement for the marker lemma, not a refinement of its constants.

The fifth point is the reason for stopping short of a full claim: the present method provably has no detector in the residual class.

5 Verification, novelty search, and limitations

- The AR approximation was checked cover-wise: closed Banach embedding, global retraction, star-refinement, locally finite partition of unity, and compactness of a finite convex hull.

- The marker induction uses exactly the uniform-tail characterization of totally bounded subsets of c_0. All five coefficient/order cases were checked independently; they cover $a_1 < b_1$ and the symmetric alternative.

- The only external mathematical inputs are the two results quoted from the source paper (SAP implies LFAP and the Banakh–Zarichnyy recognition criterion) and Josefson–Nissenzweig for the corollary.

- Bounded novelty searches (17 August 2026) used the exact source-problem phrases, "non-compact linear operator into c_0" with LFAP/SAP/AR, and "Banach envelope" with Hilbert homeomorphism/quasi-Banach. The arXiv record, the journal source and its citation neighbourhood were checked. No later full solution or this detector/envelope criterion was found. This is evidence, not a guarantee of novelty.

- The result does not settle spaces for which every continuous linear map to c_0 is compact. In particular, it does not settle the trivial-Banach-envelope quasi-Banach AR case, and it does not by itself answer the source's more general convex AR-subset problem.

Human-review recommendation. Review as a substantial candidate partial theorem. The highest-value checks are the cover-controlled retraction construction in Lemma 3, the open-cover shrinking that yields the two-adjacent partition, and the rank/coordinate bookkeeping in the five-case marker table.

References

[1] T. Banakh and R. Cauty, *Topological classification of closed convex sets in Fréchet spaces*, Studia Math. **205** (2011), 9–22; arXiv:1006.3092.

[2] A. Nissenzweig, *w^* sequential convergence*, Israel J. Math. **22** (1975), 266–272.